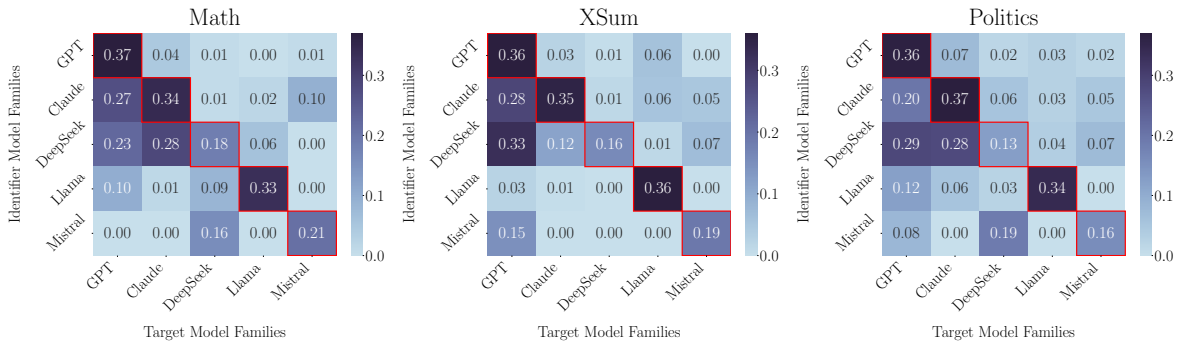


Figure 2: Heatmaps of identification **F1 scores** averaged over model family. F1 scores are consistently highest when identifier and target belong to the same family (diagonal values), indicating **strong in-family identification**. GPT models also show moderate out-of-family identifiability. See Table 6 for a qualitative example of identification. Comprehensive results are provided in Figure 9. Results in accuracy rather than F1 score are provided in Figure 11.

family's performance peaks when identifying its own members. This finding confirms the result from (Panickssery et al., 2024) but extends the scope to more models and out-of-family identifications. While most models struggle to identify out-of-family models, they show a moderate ability to identify GPT, likely because its outputs are prevalent in the training data of many other models, making its style more familiar to others.

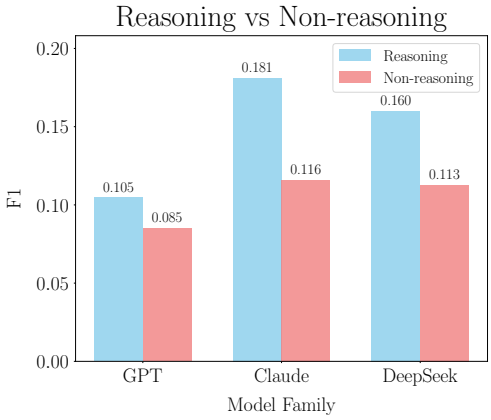


Figure 3: Identifier models with stronger reasoning capabilities achieve higher F1 scores when identifying out-of-family models.

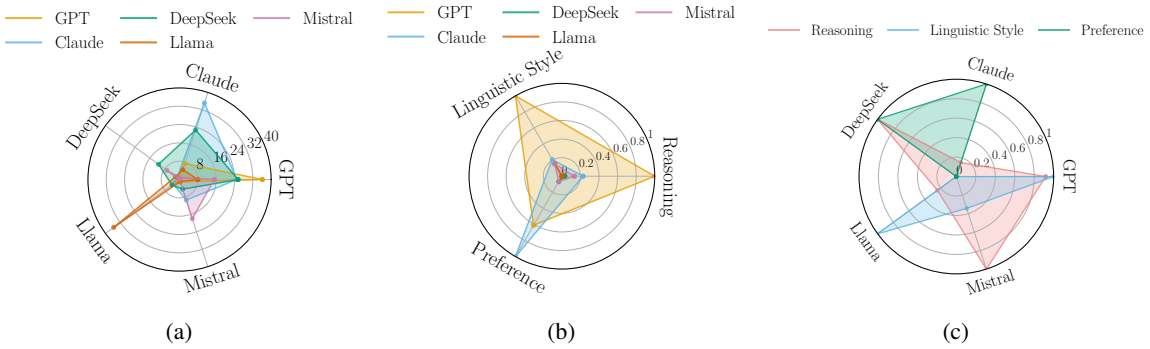
Identifier models with stronger reasoning capabilities are better at identifying out-of-family models. Given that in-family identification is a relatively straightforward task for most models (with many achieving over 90% accuracy in Figure 11), we focused on the more challenging task of out-of-family identification. We find that identifier models with stronger reasoning capabilities consistently outperform their less capable counterparts. As illustrated in Figure 3, reasoning-focused models demonstrate an improvement in F1 score of 2% to 6.5% when identifying out-of-family models. Possible explanations are: (1) reasoning models are more adept at recognizing subtle patterns, logic inconsistencies, or stylistic artifacts that serve as "fingerprints" for different model families; (2) reasoning models have more recent knowledge cutoff dates and thus contain more information of other models in their training data.

Type	Accuracy
In-range	**33.8%**
Out-of-range	14.9%

Table 1: Models show significantly higher accuracy (33.8%) when identifying targets whose data was likely included in their training set, compared to newer ones.

Familiarity through training data provides a significant advantage in identification. We hypothesized that an identifier model would be better at recognizing a target if the target model was released before the identifier's knowledge cutoff date, meaning that the target's information and outputs were part of the identifier's training data (a visualization of a relationship between models' release dates and training data cutoff dates is shown in Figure 8). Table 1 shows that identifiers obtain an 18.9% higher accuracy on average when identifying target models that were likely included in their training data compared to those that were not. This underscores that exposure to a model's output style during training is a major factor in the ability to identify it later.

Different model families are identified by distinct features in their responses. To understand what makes each model family identifiable, we analyzed which features (reasoning patterns, linguistic style, or alignment preferences) contribute most to



(a) (b) (c)

Figure 4: **(a)** F1 scores for LLM family identification. Each colored shape represents an identifier model, while each radial axis represents the target model family being identified. **(b)** Normalized F1 scores indicating which features make each target family most detectable. **(c)** Normalized F1 scores showing which response feature—reasoning patterns, linguistic style, or alignment preferences—is most characteristic for each model family.

its detection by out-of-family models. Figure 4b shows that models from the GPT family are primarily identified through their linguistic style and reasoning patterns. In contrast, Claude's expressed preferences are a more significant identifying feature compared to other models. The analysis within each family shown in Figure 4c further specifies these signatures: DeepSeek is most recognizable by its reasoning and preference patterns, Llama is dominated by its linguistic style, and for GPT, preferences contribute far less to its identifiability than its reasoning and linguistic style.

In summary, our findings suggest that while LLMs can reliably identify their own family members, recognizing others is more challenging, with some popular models being more easily identifiable, which demonstrates the emergent interlocutor awareness of LLMs. This ability is influenced by the identifier's reasoning capabilities, its familiarity with the target's outputs via training data, and the distinct fingerprints that each target family leaves in its responses across different domains. Additional experimental results and detailed analysis are provided in Appendices C and D.

4 Overview of Opportunities and Risks

Interlocutor awareness has far beyond applications than simply allowing LLMs to identify their interacting partner. One implication is the adaptive behaviors of LLMs when the identity of their interacting partner is explicitly revealed (RQ2). As each LLM possesses unique characteristics, when its conversational partners are aware of its identity, they can leverage this knowledge during interactions. This presents both opportunities and risks for the applications. To exemplify the potential impact of interlocutor awareness in detail, we present three case studies in distinctive fields:

- **Case study 1: cooperative LLM**, where a sender LLM adapts its behavior to aid the solver LLM for problem solving.
- **Case study 2: alignment risk**, where a player LLM adjusts its response to satisfy the judge LLM's preference.
- **Case study 3: safety threat**, which involves a "jailbreaker" leveraging the identified weakness of the target LLM to circumvent its safety guardrails.

We hope to use these three case studies to demonstrate the importance of understanding interlocutor awareness of LLMs and illustrate the impact of resulting behavior adaptation in fields such as multi-LLM systems, LLM alignment and LLM safety. We note that even though our evaluation results show that it is unlikely for current LLMs to recognize the targets if the targets are released after the knowledge cutoff dates, still, with more models possessing the capability of online search, they can gain the knowledge of the characteristics and capabilities of the targets which makes interlocutor awareness increasingly relevant.

5 Case Study 1: Cooperative LLMs

Motivation LLMs have been recently deployed in multi-agent settings to achieve collaborative task solving, leveraging the expertise of different models (Xiao et al., 2023). There is also a trend to enable stronger models to teach student models for fine-tuning (Lu et al., 2024; Wang et al., 2024). Interlocutor awareness enables LLMs to adapt their behaviors according to the capabilities of the inter-

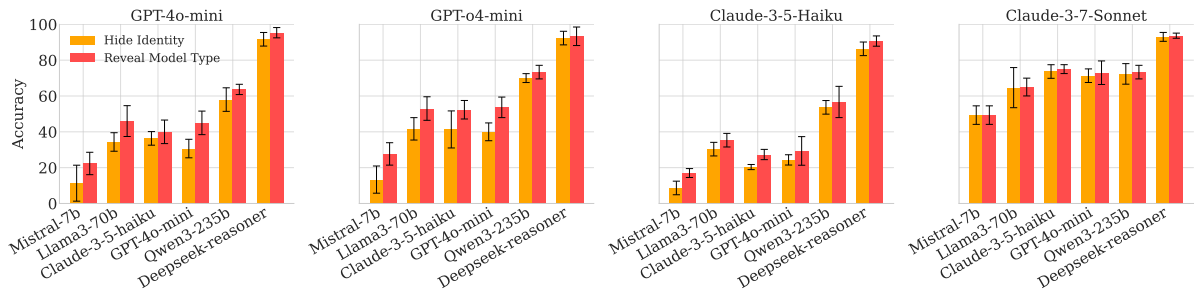


Figure 5: Averaged accuracy of the solver models on 100 randomly sampled MATH level 4 problems using the explanations generated by the sender models, denoted in the subplot title. The error bar indicates the 95% confidence interval over three independent runs. "Hide Identity" implies that the solver's identity is described as "another agent," while "Reveal Model Type" means the name of the solver denoted by the x-axis is explicitly revealed.

acting agents and thus, achieve better cooperation.

Setup We consider a cooperative mathematical problem between two LLMs. The framework consists of a "solver" and a "sender." The sender provides guidance to the solver for mathematical problems under two conditions: (a) the sender knows the solver's identity, and (b) the sender remains unaware. We assess whether the "sender" LLM tailors its explanations to the "solver" LLM's identity by using Level 4 MATH problems—chosen for their balance of challenge (they're sufficiently difficult yet still solvable by current LLMs)—as our testbed (Hendrycks et al., 2021a).

Results Figure 5 summarizes the solver accuracy across four senders and six solvers. Overall, revealing solver identity yields a consistent accuracy improvement, suggesting that senders tend to produce tailored explanations to the solvers. Especially for weaker solvers (Mistral-7B, Llama-3.3-70B), the accuracy improves by up to 10% when revealing the model name, indicating that senders are aware of the solvers' limited capabilities and provide more structured (e.g. bullet points) explanations as shown in Table 8. On the other hand, stronger solvers (Qwen-3, DeepSeek R1) exhibit negligible change, which can be attributed to their strong interpreting and reasoning ability. We also observe more noticeable accuracy improvements when senders explain their own or sibling models (e.g. o4-mini to GPT-4o-mini, Claude-3.5-Haiku to itself). This implies: (1) LLM models may draw on implicit self-knowledge to craft explanations (Laine et al., 2024); (2) LLM models may understand the explanations generated by their own or sibling models better due to the underlying patterns. The exception to our results is Claude-3.7-Sonnet, as the sender, which shows similar performance in both settings. We attribute this to its strong explanatory ability. Additional qualitative results and discussions are included in Appendix E.

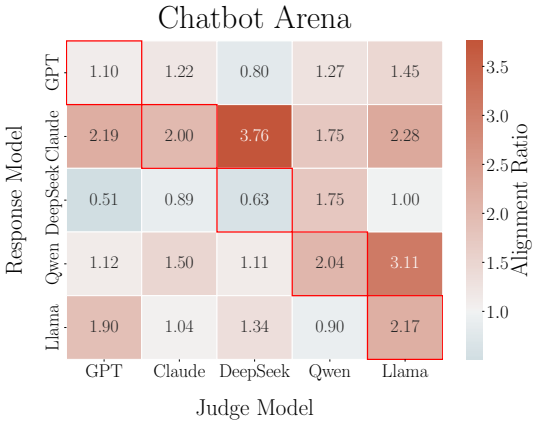


Figure 6: Win/lose ratio (identity-aware responses against identity-unaware responses) matrix illustrating strategic adaptation by responder LLMs. Each cell (i, j) denotes the win/lose ratio for responder model i when its output is judged by model j, comparing identity-aware vs. identity-unaware responses.

6 Case Study 2: Alignment Risk

Motivation As human evaluation and feedback are costly, many models are fine-tuned and aligned with feedback from another model acting as an evaluator (Casper et al., 2023). If an evaluated model is interlocutor-aware and knows the evaluator's identity, it may deliberately tailor its outputs to exploit that judge's preferences—i.e. engage in reward hacking. Such behavior undermines the validity of the evaluation and increases the risk of deceptive alignment.

Setup Inspired by Chatbot Arena frameworks (Chiang et al., 2024), we examine if revealing a